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# Variety Management

## Product Specifications Document

Version 1.0

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## Document Properties

| Item           | Name               |
|----------------|--------------------|
| Document Title | Variety Management |
| Author         | Folio3             |
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| Date           | Aug 07, 2026       |

## Review & Distribution List

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## Revision History

| Date         | Release by Name | Version | Revision Items |
|--------------|-----------------|---------|----------------|
| Aug 07, 2026 | Wania Aslam     | 1.0     |                |

## 1.1 Purpose

The purpose of this document is to define the end-to-end functional requirements for managing Variety information within AgriERP. The solution will enable Variety to be associated with Fields/Plots and Work Orders so that users can accurately select, view, filter, and track work against the relevant Variety. The document also defines the phased implementation approach for R1 and R2, including Variety usage in AgriERP and its availability in D365 F&SCM.

## 1.2 Scope

This document covers the functional handling of Variety across AgriERP Web and Mobile applications, including Work Order creation, Variety-based Field filtering, Work Order listing and details, Maps, Planning, Harvest Tickets, and D365 posting. In R1, existing Variety and acreage information will be consumed to support Work Order execution and related processes. In R2, AgriERP will provide a dedicated Plot-to-Variety mapping capability under Settings, where users can maintain Variety allocation and acreage against each Plot/Field, with validation to ensure the total Variety acreage does not exceed the Plot/Field total acreage.

## 1.3 Audience

The primary audiences for this document are Folio3 project executives, management, and wider implementation teams.

## 1.4 Functional Assumptions

These workflows will be used as a primary design artifact to confirm existing (delivered)

### Executive Summary

The Variety Management solution introduces a consistent Variety dimension across Field, Work Order, Harvest and posting processes.

# Problem Statement

The current process does not consistently maintain and use Variety across Field and Work Order levels. Users can select Fields without first narrowing them by Variety, and the Field-to-Variety relationship is not available consistently across Web, Mobile, Maps, Planning, Harvest Tickets and D365 posting. This creates a risk of selecting the wrong Field, losing Variety context during execution, and posting transactions without the same Variety information used operationally. In addition, AgriERP currently requires a controlled method to maintain Plot-level Variety mappings and acreage where multiple Varieties exist within the same Plot or Field.

# Solution Overview

The solution will be delivered in two releases. R1 will establish the GP-to-AgriERP synchronization of Field, Variety and Variety Acreage, introduce multi-select Variety on Work Order creation, filter Fields based on selected Varieties, display Variety across the required AgriERP Web and Mobile areas, support Variety-based filtering, use Variety during Harvest Ticket creation, and include Variety in D365 posting. R2 will add a Settings > Plots page in AgriERP where users can maintain one or multiple Variety mappings and acreage against each Plot, with validation that the total Variety acreage does not exceed the Plot/Field total acreage.

# Summary

The Variety Management solution will provide a consistent way to maintain and use Variety information across AgriERP operational processes. In R1, Variety and related acreage information will be available in AgriERP to support Work Order creation, multi-select Variety selection, Variety-based Field filtering, visibility across Web and Mobile screens, and posting to D365. In R2, AgriERP will introduce a dedicated Settings > Plots configuration where users can maintain Plot-to-Variety mapping and acreage allocation, with validation to ensure that the total acreage assigned to Varieties does not exceed the total Plot/Field acreage.

# Release Breakdown (R1 / R2)

| Release | Scope                                                                                                                                                                                                          | Key Outcome                                                                                           |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| R1      | GP -> AgriERP synchronization of Field, Variety and Variety Acreage; Work Order multi-select Variety; Variety-based Field filtering; Variety visibility across Web/Mobile/Maps/Planning/Harvest; D365 posting. | Variety is synchronized and used consistently throughout the operational Work Order flow and posting. |
| R2      | AgriERP Settings -> Plots page; Plot-to-Variety mapping; multiple                                                                                                                                              | Authorized users can maintain Plot-level Variety mappings and acreage within AgriERP.                 |

| Release | Scope                                                     | Key Outcome |
|---------|-----------------------------------------------------------|-------------|
|         | Varieties per Plot; acreage entry and acreage validation. |             |

# Functional Requirements - R1

## 1. GP to AgriERP Variety Synchronization

- AgriERP will receive Field, Variety and Variety Acreage information from GP.
- The integration must preserve the relationship between each Field and its mapped Variety or Varieties.
- Where a Field contains multiple Varieties, each Variety and its corresponding acreage must be available in AgriERP.
- Synchronized Variety information will be used for selection, filtering and display throughout AgriERP.
- The synchronization must associate incoming Variety information with the correct AgriERP Field using the available Field identifier/mapping.
- When the source relationship or acreage changes in GP, the synchronized AgriERP relationship must reflect the latest GP data according to the integration process.

## 2. Work Order Creation - Web

- Add a Variety field on the Work Order creation page.
- The Variety field will support multi-select so the user can select one or multiple Varieties for the same Work Order.
- Once one or more Varieties are selected, the Fields dropdown will display only Fields mapped to the selected Variety or Varieties.
- When multiple Varieties are selected, the Field list will contain the combined valid Fields mapped to the selected Variety set.
- If the Variety selection changes, any previously selected Field that is not mapped to the updated Variety selection must be removed.
- The selected Variety values must be saved against the Work Order.
- The saved Variety values must remain available to the downstream Work Order detail, listing, Mobile, Map, Harvest and posting flows.
- If multiple Varieties are selected and those Varieties are mapped to the same Field, the Variety Acreage will be summed and displayed as the total applicable acreage for that Field.

**Create New Work Order** Planned Save As Submit X

Work Order Name\*  
Enter Work Order Name

WO Start Date\* 07/31/2026 WO End Date\* 08/10/2026 Priority\* Medium

Farm\* Task Type\* Planned Task\*

Supervisor\* Responsible Inspection Templates\*

Material Template Tank Mixing Enable Tank Mixing ☒ Variety

Notes

Select Plot Task Date Range Select Range

| Select | Plot        | Customer Name | Plot Area | Operational Area | Effect |
|--------|-------------|---------------|-----------|------------------|--------|
|        | Var Non Per |               | 149       | No Record Found  | 62     |

0 Total Plots

Page Size 50 Page 1 of 13 - Showing 1 - 13

**Summary**

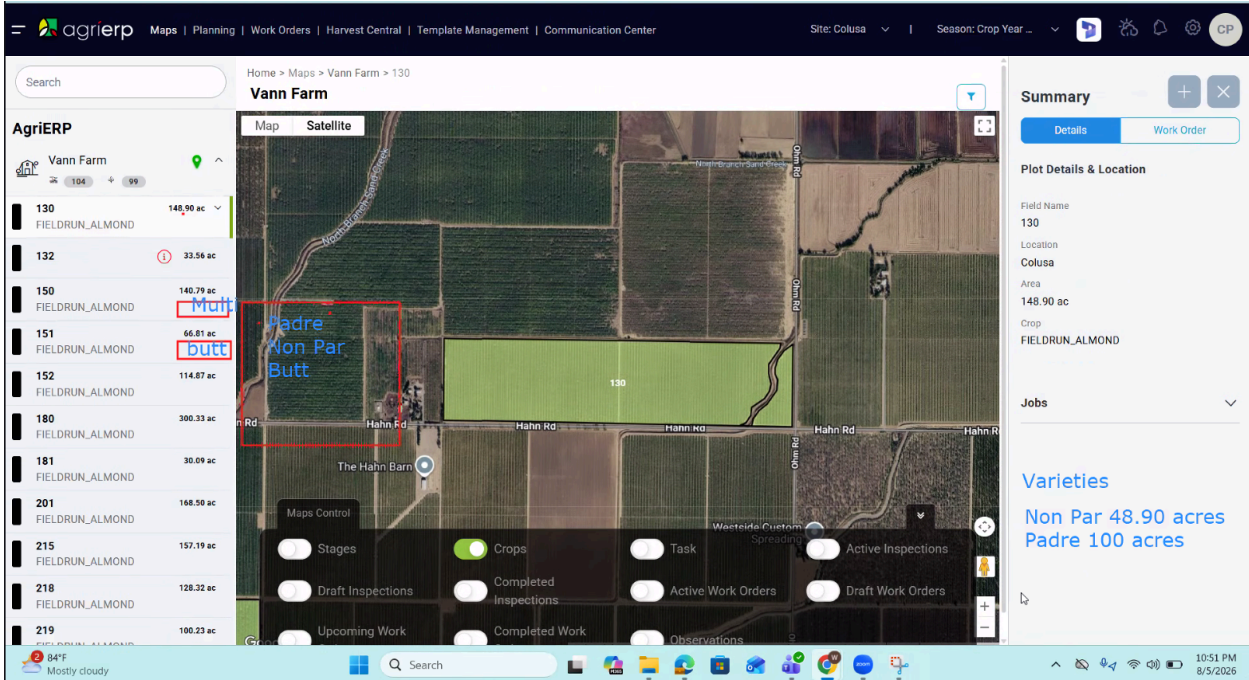
General

Plots

Resources

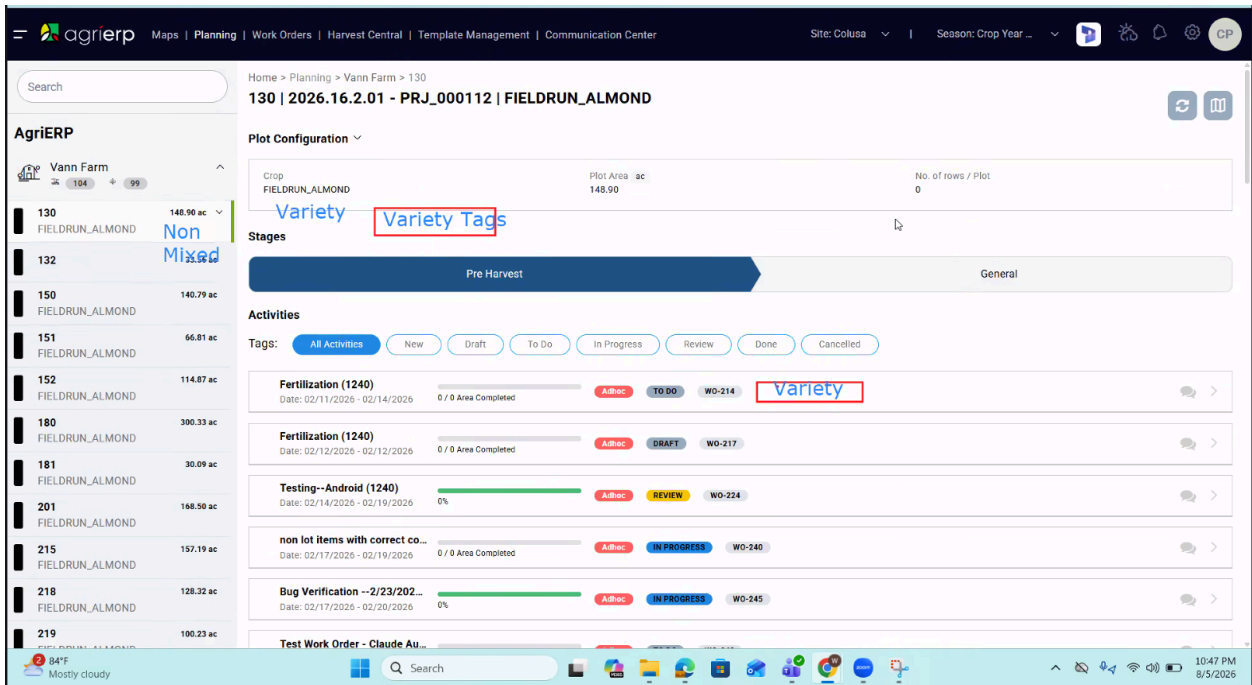
### 3. Map Screen - Web and Mobile

- When a user selects or opens a Field on the Map, display the Variety information mapped to that Field.
- Where a Field contains multiple Varieties, the Map should show the available mapped Variety context for that Field.
- Provide Variety as a Map filter so users can display Fields associated with the selected Variety.
- When a Work Order filter is applied, display only the Fields related to the selected Work Order and its applicable Variety context.
- The Variety shown on Mobile Map must match the Variety information available on the Web.



A Variety filter will be added to the Map screen on both Web and Mobile, allowing users to filter and display Fields based on the selected Variety.

#### 4. Planning - Field Configuration



On the Web Planning tab, Variety information will be displayed in the Field Summary. If a Field has up to three Varieties, all Variety names will be displayed; if it has more than three, a “Multiple Varieties” tag will be shown, and hovering over the tag will display the complete list of Varieties mapped to that Field.

## 5. Work Order Listing Page

- Add a Variety column to the Work Order listing page.
- Display the Variety or Varieties assigned to each Work Order.
- Add Variety to Advanced Filters.
- Allow users to search and filter Work Orders by Variety.
- When Variety is used as a filter, display only Work Orders mapped to that Variety.

The screenshot displays the 'Work Orders' page in the AgriERP system. The page features a table with the following columns: WO Sequence No., WO Name, Priority, Variety (highlighted with a red box), WO Start Date, WO End Date, Farm, Plots, Progress, and Task. The table contains 15 rows of work orders, all assigned to 'Vann Farm'. The 'Variety' column is currently empty for all entries. The progress bar for each work order indicates the completion status, ranging from 0% to 100%. The page also includes a top navigation bar with links to Maps, Planning, Work Orders, Harvest Central, Template Management, and Communication Center. A sidebar on the left shows the 'Work Orders' section selected. The bottom status bar displays the weather (84°F, Mostly cloudy) and the system time (10:45 PM, 8/5/2026).

| WO Sequence No. | WO Name                 | Priority | Var | WO Start Date | WO End Date | Farm      | Plots               | Progress         | Task          |
|-----------------|-------------------------|----------|-----|---------------|-------------|-----------|---------------------|------------------|---------------|
| WO-534          | Irrigation 4/29/2026    | Urgent   |     | 04/29/2026    | 05/19/2026  | Vann Farm | 338N, 321, 328, 342 | 112.801%         | Irrigation    |
| WO-1035         | Copy of RLMN Payco...   | Urgent   |     | 06/27/2026    | 07/07/2026  | Vann Farm | 315, 314            | 0% Completed     | Fertilization |
| WO-976          | RLMN Paycomm 3/7/...    | Urgent   |     | 06/27/2026    | 07/07/2026  | Vann Farm | 314, 315            | 100% Completed   | Fertilization |
| WO-975          | PLMN Paycomm 3/7/...    | Urgent   |     | 06/27/2026    | 07/07/2026  | Vann Farm | 315, 314            | 100% Completed   | Fertilization |
| WO-973          | LMN Paycomm 2/7/26...   | Urgent   |     | 06/27/2026    | 07/07/2026  | Vann Farm | 314, 315            | 70% Completed    | Fertilization |
| WO-971          | WO -Sync-WO- Save C...  | Urgent   |     | 06/27/2026    | 07/07/2026  | Vann Farm | 320                 | 50% Completed    | Fertilization |
| WO-970          | MNMN Paycomm 2/7/...    | Urgent   |     | 06/27/2026    | 07/07/2026  | Vann Farm | 315, 314            | 25.5% Completed  | Fertilization |
| WO-967          | DDDD Paycomm 2/7/2...   | Urgent   |     | 06/27/2026    | 07/07/2026  | Vann Farm | 130                 | 93.75% Completed | Fertilization |
| WO-965          | WO - R1- P1 P2 Payco... | Urgent   |     | 06/27/2026    | 07/07/2026  | Vann Farm | 289, 288            | 66.5% Completed  | Fertilization |
| WO-964          | HND Paycomm 2/7/26...   | Urgent   |     | 06/27/2026    | 07/07/2026  | Vann Farm | 130                 | 78.82% Completed | Fertilization |
| WO-951          | WO -Paycomm - 1758      | Urgent   |     | 06/26/2026    | 07/06/2026  | Vann Farm | 331, 130            | 0% Completed     | Fertilization |
| WO-925          | WO - HMD - Uptill 008   | Urgent   |     | 06/24/2026    | 07/04/2026  | Vann Farm | 316                 | 100% Completed   | Fertilization |
| WO-924          | WO HSND -020            | Urgent   |     | 06/24/2026    | 07/04/2026  | Vann Farm | 316                 | 49.68% Completed | Fertilization |

**Set Filters** [Reset] [Apply] [Close]

WO Date Range  
Select Range

Work Order Sequence No.  
All

Work Order Name  
Work Order Name

Farm  
All

Task  
All

Priority  
All

Created By  
All

Submitted By  
All

Resource  
All

Variety

## 6. Work Order Detail Page

- Add a Variety column under the Fields tab on the Work Order Detail page.
- Display the mapped Variety against each Field.
- Where multiple Varieties apply, preserve the Field-to-Variety relationship so users can identify the correct Variety context for each Field.

The screenshot displays the AgriERP interface for a Work Order (WO-1035). The top navigation bar includes links for Maps, Planning, Work Orders, Harvest Central, Template Management, and Communication Center. The main header shows the work order details: WO-1035 | Copy of RLMN Paycomm 3/7/26 First, with an 'In Progress' status and an 'End Work Order' button.

Below the header, there are several sections:
 

- Start Date:** 06/27/2026
- End Date:** 07/07/2026
- Progress:** 0% Completed
- Supervisor:** Abdul Wahab (View Spent Hours)
- Farm:** Varin Farm
- Work Order Type:** Planned
- Task Type:** Planned
- Task:** Fertilization
- Crop Stage:** Pre Harvest
- Assets:** 0 Assets, 0 Resources
- Material Template:** N/A
- Created By:** Crop Planner
- Submitted By:** Crop Planner
- Responsible:** N/A
- Inspection Template:** N/A
- Notes:** N/A
- Approver Comments:** N/A

The **Plots** section contains a table with the following columns: Plots, Customer Name, Progress, Crop Variety, Total Area - ac, Operational Area - ac, Effective Area, Spent Hours, and Attachments. Two plots are listed:
 

- Plot 315 (PRJ\_000062):** Progress 0 / 25 Area Completed, Crop Variety FIELDRUN\_ALMOND, Total Area 21.65, Operational Area 21.65, Effective Area 20, Spent Hours 0 / 0.
- Plot 314 (PRJ\_000061):** Progress 0 / 75 Area Completed, Crop Variety FIELDRUN\_ALMOND, Total Area 76.12, Operational Area 76.12, Effective Area 75, Spent Hours 0 / 0.

The **Inputs** section shows a table with columns: Name, Rate, Received Qty/Per Total Qty, Consumed Qty / Total Qty, and Action. Below the table, there are status indicators: Total Budgeted, In Progress (not yet complete), 100% Complete, and Overdue.

A red box highlights the 'Variety' tag in the 'Crop Variety' column of the first plot. The tag is labeled 'Variety' and is linked to the 'FIELDRUN\_ALMOND' variety.

## 7. Mobile Work Order

- When a Work Order is opened on Mobile, display Variety on each Field card.
- If a Field has multiple Varieties, up to two Variety tags will be displayed directly on the Field. If more than two Varieties are mapped, the remaining count will be shown as a "+N" tag (for example, +2), and tapping the tag will open a bottom sheet displaying the complete list of Varieties mapped to that Field.
- Variety information will also be displayed on the Mobile Work Order Summary tab, showing the Variety/Varieties associated with the selected Field and Work Order.

## 8. D365 Posting

- Include the applicable Variety when the Work Order or related transaction is posted to D365.
- Variety must be available against the relevant D365 posting transaction or journal line.

# Functional Requirements - R2

## 1. Settings - Plots Page

- Add a Plots page under AgriERP Settings.
- The page will display all available Plots in AgriERP.
- Each Plot should display sufficient identifying context such as Plot name/ID, related Field/Farm and total acreage where available.
- Authorized users will be able to open a Plot and maintain its Variety mapping.

## 2. Plot-to-Variety Mapping

- Allow one or multiple Varieties to be mapped against a Plot.
- Provide an acreage value against each mapped Variety.
- Allow users to add additional Variety mappings when more than one Variety exists within the same Plot.
- Allow authorized users to update or remove a mapping subject to the applicable business permissions and usage rules.
- Saved Plot-to-Variety mappings will be available for AgriERP filtering, display and operational selection.

## 3. Acreage Allocation

- Display the Plot/Field total acreage as the maximum available acreage for Variety allocation.
- Capture acreage against each Variety mapped to the Plot.
- Calculate the total acreage assigned across all mapped Varieties.
- Prevent saving when the combined Variety acreage exceeds the Plot/Field total acreage.
- The combined Variety acreage may be less than the Plot/Field total acreage where the business has not allocated the complete area.

## 4. Use of R2 Mapping in Operational Screens

- The current valid Plot-to-Variety mapping maintained in R2 will be available to AgriERP operational screens that depend on Variety-based Field/Plot filtering and display.
- Work Order, Maps, Planning, Detail and Mobile experiences should use the valid AgriERP mapping when determining the Variety associated with a Plot.
- A reverse update from R2 configuration back to GP is not part of the confirmed requirement and requires separate confirmation if needed.

# Integrations

## Integration with Grower Portal (GP)

In R1, GP will provide the Field, Variety and Variety Acreage relationship required by AgriERP. AgriERP will synchronize this information and use it as the basis for Variety selection, Field filtering, acreage visibility and downstream operational screens. The existing Harvest specification already identifies GP as the source of Variety reference data for Harvest processing; the R1 Variety solution extends that concept by bringing the Field-to-Variety and Variety Acreage relationship into the broader Work Order process.

## Integration with D365 F&SCM

In R1, Variety information required for downstream processing will be available in D365. When AgriERP posts the Work Order or applicable transaction, the system will send the applicable Variety with the relevant Field/Work Order context so the same Variety used operationally is available on the D365 posting transaction or journal line. If Variety is required for a posting and is missing, posting will be restricted until the Variety information is available.

## R2 Configuration

In R2, the Field-to-Variety mapping maintained in AgriERP will be synchronized from AgriERP to ERP, so the corresponding Field and Variety information is available in ERP for downstream processing and posting.

## App User

| App Access       | Roles                      | Description                                                                                                                                                                                                  |
|------------------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Farm Web App     | Admin / Manager            | Create and review Work Orders, select one or multiple Varieties, select filtered Fields, review Variety on listing/detail/Maps/Planning, create Harvest Tickets, and post transactions with Variety to D365. |
| Farm Mobile App  | Supervisor / Field User    | Open assigned Work Orders, review Variety on Field cards and Map, and use Variety during Harvest Ticket creation according to the Work Order/Field context.                                                  |
| AgriERP Settings | Authorized Admin / Manager | In R2, open the Plots page and maintain Plot-to-Variety mappings and Variety acreage within the allowed total acreage.                                                                                       |

## Persona | Farm Operations Manager - Web App

The Farm Operations Manager creates and reviews Work Orders and needs to select the correct Fields for the intended Variety. Variety-based filtering reduces incorrect Field selection, while listing, detail, Map and Planning visibility allows the manager to verify the same Variety throughout the Work Order lifecycle. The manager also requires the same Variety context to be sent to D365 during posting.

## Persona | Field Supervisor - Mobile App

The Field Supervisor executes Work Orders in the field and needs clear visibility of the Variety associated with each Field. Mobile should display the same Variety information maintained on Web so the supervisor can confirm the correct Field and Variety during execution and Harvest Ticket creation.

## Farm Operations Manager User Journey

- The user logs in to AgriERP Web and opens Work Order creation.

- The user selects one or multiple Varieties.
- The system filters the available Fields based on the selected Varieties.
- The user selects the required Fields and completes the remaining Work Order information.
- The user submits the Work Order and the selected Variety information is saved.
- The user can review Variety from Work Order listing, detail, Maps, Planning and Advanced Filters.
- When the Work Order or related transaction is posted, the applicable Variety is sent to D365.

## Field Supervisor User Journey

- The user logs in to AgriERP Mobile and opens an assigned Work Order.
- Mobile displays Variety against each applicable Field card.
- The user can open the Map and review the Field with its Variety context.
- When creating a Harvest Ticket, the user selects Variety and the Field list is filtered accordingly.
- The saved Harvest Ticket retains the selected Variety for downstream processing.

## R2 Configuration User Journey

- An authorized user opens Settings > Plots.
- The system displays all available Plots with identifying Field/Farm information and total acreage where available.
- The user opens the required Plot and adds one or multiple Variety mappings.
- The user enters acreage against each mapped Variety.
- The system calculates total mapped Variety acreage and validates it against Plot/Field total acreage.
- The user saves the configuration and the mapping becomes available to AgriERP operational filtering and display.

# Business Cases

1. Users can identify the correct Fields for a Work Order by selecting Variety first, reducing incorrect Field selection.
2. A single Work Order can support multiple Varieties where operational work covers more than one Variety.
3. Fields containing multiple Varieties can retain Variety-specific acreage.
4. Farm Operations Managers can search and review Work Orders by Variety without opening each Work Order individually.
5. Field Supervisors can confirm the expected Variety from Mobile before or during field execution.
6. Harvest Ticket creation can use Variety to reduce selection of an incorrect Field.
7. D365 receives the same Variety context used operationally in AgriERP, improving consistency between execution and ERP posting.
8. R2 provides a controlled AgriERP Settings page to maintain Plot-to-Variety relationships when operational mapping is required.
9. Acreage validation prevents the sum of Variety allocations from exceeding the physical Plot/Field acreage.

# Data Dictionary

| Field Name            | Description                                                                  | Source                           | Data Type           | Release |
|-----------------------|------------------------------------------------------------------------------|----------------------------------|---------------------|---------|
| Variety Acreage       | Acreage associated with a Variety within the relevant Field/Plot.            | GP in R1; AgriERP Settings in R2 | Decimal             | R1 / R2 |
| Work Order Variety    | One or multiple Varieties selected during Work Order creation.               | AgriERP Web                      | Multi-select Lookup | R1      |
| Field-Variety Mapping | Relationship used to filter Fields according to selected Variety.            | GP -> AgriERP                    | Reference           | R1      |
| D365 Variety          | Variety reference posted with the applicable Work Order/transaction context. | AgriERP -> D365                  | Reference           | R1      |